



A
Project Synopsis
on
Voice-To-Post
(AI-Driven Voice-Based Social Media Post Generator)
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ABSTRACT

Voice-To-Post is an AI-powered mobile application designed to generate optimized, safe, and platform-ready social media posts using voice commands. The system integrates mobile engineering principles for a smooth, lightweight Android experience, and machine learning engineering methods for retrieval, generation, and safety filtering.

The Android frontend captures the user's speech and communicates with a cloud-based backend. All heavy ML operations—including speech recognition, semantic retrieval, RAG-based post generation, vector search (FAISS), and confidence scoring—are executed on the server. A similarity and safety threshold ensures that generated posts remain factual, grounded, and safe for public posting.

This end-to-end architecture enables users to quickly create meaningful, trend-aligned content without manual writing or trend research, offering a seamless blend of mobile usability and ML reliability.

INTRODUCTION

Social media users often struggle to generate timely, trend-aware, and safe content. With dynamic online environments, manual content creation becomes time-consuming and inconsistent. Voice-To-Post aims to solve this through a voice-first interface paired with a robust backend ML pipeline.

Motivation

- Difficulty drafting platform-specific posts.
- High hallucination risk in unguided LLM outputs.
- Variation in tone, length, and structure across platforms.
- Rising demand for hands-free, voice-driven content creation.

Gaps Identified

- Lack of real-time, retrieval-aware post generation.
- Limited systems integrating speech → retrieval → generation → safety.
- Absence of confidence-thresholding in common tools.
- Heavy ML models cannot run efficiently on mobile hardware.
- Weak separation of UI and backend intelligence in existing tools.

Our Contribution

- A complete **voice** → **text** → **retrieval** → **generation** → **safety** → **publish** pipeline.
- FAISS-based semantic retrieval for factual grounding.
- Multi-factor confidence scoring to eliminate hallucination.
- Lightweight Android app using Jetpack Compose.
- Mobile-offloading cloud architecture for smooth performance.
- Platform-aware optimization for LinkedIn, X, and Instagram.

LITERATURE SURVEY

Traditional LLM systems produce fluent text but often lack factual grounding. Retrieval-Augmented Generation (RAG) improves accuracy through integration of external knowledge sources. FAISS (Meta AI) is widely used for scalable vector similarity search.

Existing voice-based tools primarily focus on transcription, not content generation. Common social media content tools lack:

- contextual retrieval,
- multi-factor safety scoring,
- voice-based content triggers,
- mobile-cloud ML offloading mechanisms.

Mobile AI research strongly suggests delegating resource-intensive tasks to cloud servers while maintaining a responsive mobile UI. Voice-To-Post aligns well with these principles.

METHODOLOGY / PLANNING OF WORK

The pipeline integrates mobile engineering and ML engineering:

1. **Voice Input (Mobile App)**
User speech is captured and securely transmitted to the backend.
2. **Speech-to-Text (Backend ML)**
Whisper/Deepgram/Google STT transcribes the input.
3. **Context Retrieval using FAISS**
 - a. Embeddings of trends/news stored in a vector database.
 - b. User query is embedded and matched using cosine similarity.
 - c. Low similarity → reject output to avoid hallucinations.
4. **RAG-Based Post Generation**
 - a. Retrieved documents + query → LLM output.
 - b. Platform-optimized formatting.
5. **Confidence & Safety Filtering**
Evaluates AI confidence, retrieval relevance, safety score, and engagement potential.
6. **Response to Mobile App**
Final optimized post + confidence score shown to user.
7. **Optional Publishing (Future Scope)**
Auto-posting through LinkedIn/X APIs.

FACILITIES REQUIRED

Software

- Android Studio (Kotlin + Jetpack Compose)
- Python (FastAPI / Flask)
- FAISS vector search
- LangChain RAG pipeline
- Whisper / Deepgram / Google STT
- PyTorch / TensorFlow

Hardware

- Android smartphone with good microphone
- Cloud GPU backend
- 8–16 GB RAM server
- Vector DB storage for embeddings

IMPROVEMENTS TO METHODOLOGY & EVALUATION PLAN

Planned Enhancements

- Adaptive similarity thresholds per platform.
- User-feedback-based retrieval reinforcement.
- Human-in-the-loop safety calibration.
- A/B testing for template quality.

Evaluation Metrics

- Precision/Recall for retrieval
- BLEU/ROUGE for generation

- User satisfaction scoring
- Toxicity FPR/FNR

TABLES

Table 1: Safety Metrics & Confidence Scoring

Component	Details
Safety Detection	Toxicity, harassment, hate, harmful content
Datasets	Jigsaw Toxicity, Wiki Toxicity, OpenAI Moderation
Confidence Formula	$C = w1(\text{AI Confidence}) + w2(\text{Retrieval Relevance}) + w3(\text{Safety Score}) + w4(\text{Engagement Potential})$
Threshold	Output accepted if $C \geq T$

Table 2: Recommended Datasets & APIs

Category	Examples
Speech Datasets	Mozilla Common Voice, LibriSpeech, VoxCeleb
News APIs	NewsAPI, GDELT, EventRegistry
STT APIs	Whisper, Google STT, Deepgram
Safety Datasets	Jigsaw Toxicity, Wiki Toxicity, OpenAI Moderation

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